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$$\begin{aligned} & \int \frac{dx}{\sqrt{3+2x-x^2}} \\ &= \int \frac{dx}{\sqrt{-(x^2-2x+1-4)}} \\ &= \int \frac{dx}{\sqrt{-(x^2-2x+1-4)}} \\ &= \int \frac{dx}{\sqrt{-((x-1)^2-4)}} \\ &= \int \frac{dx}{\sqrt{4-(x-1)^2}} \\ &= \int \frac{du}{\sqrt{4-u^2}} \\ &= \int \frac{du}{\sqrt{2^2-u^2}} \\ &= \text{Bgsin} \frac{u}{2} + c \\ &= \text{Bgsin} \frac{x-1}{2} + c \end{aligned}$$

References